

Chapter 1: Nature of Matter

Introduction

Matter is anything that has mass and occupies space. Every object around us, from air to mountains, is made of matter. Matter exists in three common physical states: solid, liquid, and gas. The behavior of matter depends on the arrangement and movement of its particles.

Scientists study matter to understand physical and chemical changes occurring in nature. Knowledge of matter forms the foundation of chemistry because every chemical reaction involves changes in matter.

Characteristics of Matter

Matter possesses several important characteristics.

- It has mass.
- It occupies space.
- It is made up of tiny particles.
- The particles are continuously moving.
- There is space between particles.
- Particles attract each other.

States of Matter

Solid

A solid has a definite shape and definite volume. The particles are closely packed together and vibrate around fixed positions.

Examples:

- Iron
- Ice
- Wood
- Gold

Properties

- Fixed shape
- High density
- Very little compressibility
- Strong intermolecular forces

Liquid

Liquids have a fixed volume but no fixed shape.

Examples

- Water
- Milk
- Oil
- Alcohol

Properties

- Flow easily
- Moderate intermolecular forces
- Slightly compressible

Gas

Gases have neither definite shape nor definite volume.

Examples

- Oxygen

Nitrogen
Carbon dioxide

Properties

Highly compressible
Low density
Large spaces between particles
Particle Theory

The kinetic molecular theory explains the behavior of matter.

According to this theory

Matter consists of tiny particles.
Particles are always moving.
Higher temperature means faster particle movement.
Attractive forces exist between particles.
Physical Changes

A physical change does not produce a new substance.

Examples

Melting ice
Boiling water
Breaking glass
Chemical Changes

Chemical changes produce new substances.

Examples

Rusting
Burning paper
Digestion
Cooking food
Real-Life Applications

Understanding matter helps in

Manufacturing industries
Food preservation
Medicine
Construction
Environmental science
Worked Example

Question

Why does ice float on water?

Answer

Ice has a lower density than liquid water because water molecules form an open crystal structure during freezing.

Practice Questions

Define matter.
Explain three states of matter.
Differentiate solids and liquids.
Explain physical change.
Explain chemical change.
MCQs

1. Matter occupies

A. Time

B. Space

C. Energy

D. Heat

Answer: B

2. Which state has fixed shape?

A. Gas

B. Liquid

C. Solid

D. Plasma

Answer: C

Chapter 2: Atomic Structure

Introduction

Every substance consists of atoms. An atom is the smallest particle of an element that retains its chemical properties. Modern atomic theory explains atomic structure using subatomic particles including protons, neutrons, and electrons.

Discovery of the Atom

Ancient philosophers proposed that matter is made of indivisible particles.

Later scientists developed atomic theory through experiments.

Important contributors include

John Dalton

J. J. Thomson

Ernest Rutherford

Niels Bohr

James Chadwick

Structure of an Atom

Atoms contain three fundamental particles.

Proton

Positive charge

Located inside nucleus

Mass approximately 1 amu

Neutron

No charge

Located inside nucleus

Mass approximately 1 amu

Electron

Negative charge

Revolves around nucleus

Very small mass

Atomic Number

Atomic number represents the number of protons.

Example

Carbon

Atomic Number = 6

Mass Number

Mass Number

= Protons + Neutrons

Example

Protons = 6

Neutrons = 8

Mass Number = 14

Isotopes

Atoms of the same element with different numbers of neutrons.

Examples

Carbon-12

Carbon-13

Carbon-14

Applications

Medical diagnosis

Archaeology

Nuclear power

Electron Configuration

Electrons occupy shells.

K Shell → 2 electrons

L Shell → 8 electrons

M Shell → 18 electrons

Bohr Model

Bohr proposed that electrons revolve in fixed energy levels.

Electrons absorb or release energy when moving between energy levels.

Applications

Lasers

Medical imaging

Nuclear medicine

Electronics

Spectroscopy

Worked Example

Question

Calculate the number of neutrons in Sodium-23.

Solution

Mass Number = 23

Atomic Number = 11

Neutrons = 23 - 11

= 12

Practice Questions

Define atom.

Differentiate proton and electron.

What is atomic number?

Explain isotopes.

Draw Bohr model.

MCQs

1. Electron carries

A. Positive charge

B. Negative charge

C. Neutral charge

D. No mass

Answer: B

2. Nucleus contains

A. Electrons only

B. Protons and neutrons

C. Electrons and neutrons

D. None

Answer: B

Chapter 3: Periodic Table

Introduction

The periodic table is a systematic arrangement of all known chemical elements based on their atomic numbers. It helps scientists predict the physical and chemical properties of elements and understand relationships between them. The modern periodic table is one of the most important tools in chemistry because it organizes elements into groups and periods according to similar characteristics.

The periodic table contains over 118 known elements, each represented by a unique chemical symbol. Elements in the same group generally exhibit similar chemical behavior because they have the same number of valence electrons.

History of the Periodic Table

The development of the periodic table involved contributions from several scientists.

Johann Döbereiner
Proposed the Law of Triads.
Grouped elements with similar properties into sets of three.

Example

Lithium
Sodium
Potassium
John Newlands

Proposed the Law of Octaves.

According to this law, every eighth element has similar properties.

Although successful for lighter elements, it failed for heavier elements.

Dmitri Mendeleev

Mendeleev arranged elements according to increasing atomic mass.

His achievements include:

Left gaps for undiscovered elements.
Predicted properties of unknown elements.
Corrected atomic masses of several elements.

His periodic table became the foundation of the modern periodic table.

Henry Moseley

Henry Moseley arranged elements according to increasing atomic number rather than atomic mass.

This solved inconsistencies found in Mendeleev's table.

The modern periodic table is based on Moseley's work.

Structure of the Modern Periodic Table

The periodic table consists of

7 Periods (Horizontal rows)
18 Groups (Vertical columns)

Each element occupies a unique position based on its atomic number.

Periods

Periods are horizontal rows.

Properties of periods

Atomic number increases from left to right.
Number of electron shells increases.
Metallic character decreases.
Non-metallic character increases.

Example

Period 2

Lithium

Beryllium
Boron
Carbon
Nitrogen
Oxygen
Fluorine
Neon
Groups

Groups are vertical columns.

Elements within the same group have

Similar chemical properties
Same number of valence electrons
Similar oxidation states

Example

Group 1

Lithium
Sodium
Potassium
Rubidium
Cesium

These are called Alkali Metals.

Classification of Elements

The periodic table classifies elements into different categories.

Metals

Properties

Good conductors
Lustrous
Malleable
Ductile
High melting point

Examples

Iron
Copper
Aluminum
Gold

Applications

Construction
Electrical wiring
Vehicles
Machinery
Non-metals

Properties

Poor conductors
Brittle
Low density
Not malleable

Examples

Oxygen
Nitrogen
Sulfur
Chlorine

Applications

Fertilizers
Medicines
Water purification
Metalloids

Metalloids possess properties of both metals and non-metals.

Examples

Silicon
Germanium
Arsenic

Applications

Semiconductors
Solar panels
Computer chips
Blocks of the Periodic Table

Elements are divided into four blocks.

s-block

Contains Groups 1 and 2.

Characteristics

Highly reactive
Soft metals
Form ionic compounds

Examples

Sodium
Magnesium
p-block

Contains Groups 13-18.

Includes

Metals
Non-metals
Noble gases

Examples

Carbon
Nitrogen
Oxygen
Neon
d-block

Known as Transition Metals.

Examples

Iron
Copper
Zinc
Nickel

Properties

Variable oxidation states
Colored compounds
Catalytic activity

Applications

Steel manufacturing
Coins
Industrial catalysts
f-block

Includes

Lanthanides
Actinides

Applications

Nuclear energy
Medical imaging
Magnets
Periodic Trends
Atomic Radius

Atomic radius is the distance from the nucleus to the outermost electron shell.

Trend

Decreases across a period.
Increases down a group.
Ionization Energy

Energy required to remove an electron from an atom.

Trend

Increases across a period.
Decreases down a group.
Electronegativity

Ability of an atom to attract electrons.

Trend

Increases across a period.
Decreases down a group.

Fluorine has the highest electronegativity.

Metallic Character

Trend

Decreases across a period.
Increases down a group.
Noble Gases

Group 18 elements are called noble gases.

Examples

Helium
Neon
Argon
Krypton
Xenon

Properties

Chemically inert
Colorless
Odorless
Stable electron configuration

Applications

Neon signs
Balloons
Welding
Lighting
Importance of the Periodic Table

The periodic table helps scientists

Predict properties of unknown elements.
Understand chemical reactions.
Design new materials.
Study atomic structure.
Discover new elements.
Develop medicines.
Manufacture industrial chemicals.
Real-Life Applications

The periodic table is used in

Medical research
Electronics
Aerospace engineering
Agriculture
Environmental science
Battery technology
Semiconductor manufacturing
Worked Example
Question

Identify the group and period of Sodium (Na).

Solution

Atomic Number = 11

Electron Configuration

2,8,1

Therefore

Group = 1
Period = 3

Sodium is an Alkali Metal.

Practice Questions

What is the periodic table?

State the modern periodic law.

Differentiate groups and periods.

Explain atomic radius.

Define ionization energy.

What are noble gases?

Explain the importance of Mendeleev's periodic table.

Compare metals and non-metals.

What are metalloids?

Explain periodic trends with examples.

Multiple Choice Questions

1. The modern periodic table is arranged according to

A. Atomic Mass

B. Atomic Number

C. Density

D. Volume

Answer: B

2. How many groups are present in the modern periodic table?

A. 7

B. 8

C. 18

D. 20

Answer: C

3. Which group contains noble gases?

A. Group 1

B. Group 2

C. Group 17

D. Group 18

Answer: D

4. Which element has the highest electronegativity?

A. Oxygen

B. Chlorine

C. Fluorine

D. Nitrogen

Answer: C

5. Which block contains transition metals?

- A. s-block
- B. p-block
- C. d-block
- D. f-block

Answer: C

6. Which scientist developed the modern periodic law?

- A. Dalton
- B. Moseley
- C. Rutherford
- D. Bohr

Answer: B

7. Which element belongs to Group 1?

- A. Chlorine
- B. Sodium
- C. Oxygen
- D. Neon

Answer: B

8. Which of the following is a metalloid?

- A. Iron
- B. Silicon
- C. Oxygen
- D. Sodium

Answer: B

9. Atomic radius generally _____ across a period.

- A. Increases
- B. Decreases
- C. Remains constant
- D. Doubles

Answer: B

10. Which of the following is a noble gas?

- A. Hydrogen
- B. Carbon
- C. Argon
- D. Sulfur

Answer: C

Chapter Summary

In this chapter, you learned about the development and structure of the modern periodic table, including the contributions of Döbereiner, Newlands, Mendeleev, and Moseley. You studied groups, periods, metals, non-metals, metalloids, and the s-, p-, d-, and f-blocks. You also explored important periodic trends such as atomic radius, ionization energy, electronegativity, and metallic character. The chapter concluded with practical applications, worked examples, revision questions, and MCQs to reinforce understanding.